# Osama Atta

## Senior Full-Stack Software Engineer

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## Professional Summary

Senior full-stack engineer with five years at Union Pacific building and modernizing enterprise Java applications with Jakarta EE, Spring/Spring Boot, and Maven. Experienced in REST API development, framework and platform migrations, Kubernetes deployments, and production on-call support at 99.94% uptime. Built an AI-powered payment extraction app and evaluated spec-driven AI development (OpenSpec) for team adoption, backed by an AI certificate in progress.

## Skills

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| **Java & Frameworks** | Java \| Jakarta EE \| Spring \| Spring Boot \| JMS \| JDBC \| SQL \| TypeScript (Angular) \| HTML/CSS \| Python |
| **Build & Platform** | Maven (parent POMs, BOMs, plugins) \| Kubernetes \| Linux/Unix \| Microservices \| REST APIs |
| **Quality & DevOps** | JUnit \| Mockito \| TestNG \| Jasmine \| Git \| GitHub Actions \| Jenkins \| CI/CD \| Liquibase \| Agile |
| **AI** | AI-Assisted Development \| Spec-Driven Development (OpenSpec) \| PyTorch |

## Experience

### Union Pacific Railroad — **Oct 2021 - Present | Omaha, NE**

***Senior Project Engineer - FinTech*** — *Dec 2025 - Present*

- Modernized accounts receivable and payable applications by upgrading Java 8 to Java 17 and Spring Boot, migrating javax to Jakarta EE, and replacing JMS messaging with HTTP service calls.
- Reduced security risk by remediating CVEs through dependency and framework upgrades, and cleaned up parent POMs to simplify Maven dependency management.
- Built an AI-powered Java/Spring application that extracts check payment data from customer emails and applies it to accounts receivable, improving read accuracy by 5% over the legacy system and flagging exceptions for manual review.
- Built a full-stack Java/Spring and TypeScript/Angular application with RESTful APIs and an Oracle back-end, giving the marketing team a validated, self-service way to maintain marketing dimensions previously handled through manual requests.
- Evaluated OpenSpec, a spec-driven development framework for AI coding agents, defining requirements and resolving open questions before agent-driven implementation; presenting adoption recommendations to other engineering teams.
- Partnered with internal customers to build and enhance a payroll workflow application for approving off-cycle payouts.

***Project Engineer - Operations*** — *Oct 2024 - Dec 2025*

- Led migration from GitLab to GitHub and Jenkins to GitHub Actions, resolving integration issues to keep releases stable.
- Led a team of 4 interns in rewriting two high-usage web applications from Apache Wicket to responsive Angular.
- Built 2 web applications using an Oracle database, Java/Spring back-end, and TypeScript/Angular front-end to track crew productivity metrics and improve locomotive utilization, achieving $4.6 million in savings.
- Shipped customer-requested features and critical fixes for Android and web apps, cutting tickets 63% year over year.
- Helped roll out AI developer tools later adopted company-wide, providing feedback to optimize integration.

***Associate Senior Application Developer - Operations*** — *May 2023 - Oct 2024*

- Expanded telemetry data collection from 10,000+ Android devices by developing Java-based microservices and optimizing back-end data processing.
- Represented the Mobile Work Order team in Union Pacific's historic NetControl modernization, supporting enterprise cutover from legacy mainframes to an API-driven platform.
- Achieved 99.94% uptime for multiple products by providing on-call support through proactive monitoring and rapid fixes.
- Increased web application usage by 15% by spearheading a UX improvement initiative and mentoring a UX intern through research, prototyping, and user testing.

***Technology Year-Round Intern - Operations*** — *Oct 2021 - May 2023*

- Upgraded a JMS-based Enterprise Event Bus to a leaner architecture using fewer queues for low-volume events.
- Developed and maintained a mobile work order application using Java/Spring and TypeScript/Angular, maintaining 80% test coverage with JUnit, Mockito, and Jasmine.
- Improved maintainability and performance by migrating a legacy AngularJS application to Angular 11.

## Education

**Artificial Intelligence Certificate** — **Expected December 2026**
*University of Nebraska at Omaha* — *Omaha, NE*

**Bachelor of Science - Computer Engineering** — **May 2023**
*University of Nebraska - Lincoln* — *Lincoln, NE*
Minors: Mathematics, Computer Science

## Notable Projects

### Thermal Water Storage System — **Oct 2022 - May 2023**

- Built an IoT thermal water storage system with STM32/ESP8266 firmware, MQTT messaging, and Node-RED/MySQL analytics.